

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 23

INFORMATION & COMMUNICATION TECHNOLOGY (ICT) - 4

INTERNET OF THINGS (IOT)

- It means taking all the things in the world and connecting them to the internet.
- When something is connected to the internet, it can send information or receive information, or both. This ability to send or receive information makes things smart.
- To be smart, a device doesn't need to have super storage or a super computer inside of it. All a device has to do is connect to super storage or to a super computer.

WHY IOT IS VERY IMPORTANT?

Example: Right now we can listen to just about any song in the world, not because our phone has every song in the world stored on it. It's because every song in the world is stored somewhere else, but our phone can send information (asking for that song) and then receive information (streaming that song on our phone).



INTERNET OF THINGS (IOT)

- Things that collect information and then send it.
- Things that receive information and then act on it.
- Things that do both.



INTERNET OF THINGS (IOT)

1 COLLECTING AND SENDING INFORMATION

- This means sensors. Sensors could be temperature sensors, motion sensors, moisture sensors, air quality sensors, light sensors etc.
- These sensors allow us to collect information from the environment which, in turn, allows us to make more intelligent decisions.
 - ✓ **Example:** On the farm, automatically getting information about the soil moisture can tell farmers exactly when their crops need to be watered. Instead of watering too much or watering too little the farmer can ensure that crops get exactly the right amount of water.



SOIL MOISTURE SENSOR

INTERNET OF THINGS (IOT)

2 RECEIVING AND ACTING ON INFORMATION

- We're all very familiar with machines getting information and then acting. A printer receives a document and it prints it. A car receives a signal from the car keys and the doors open, etc.
- The real power of the Internet of Things arises when things can do both of the above. Things that collect information and send it, but also receive information and act on it.



INTERNET OF THINGS (IOT)

3 SENDING ... RECEIVING ... DOING BOTH

- Let us go back to the farming example. The sensors can collect information about the soil moisture to tell the farmer how much to water the crops, but we don't actually need the farmer. Instead, the irrigation system can automatically turn on as needed, based on how much moisture is in the soil.
- If the irrigation system receives information about the weather from its internet connection, it can also know when it's going to rain and decide not to water the crops, because they will be watered by the rain.



INTERNET OF THINGS (IOT)

3 SENDING ... RECEIVING ... DOING BOTH

- All this information about the soil moisture, how much the irrigation system is watering the crops, and how well the crops actually grow can be collected and sent to supercomputers that run amazing algorithms that can make sense of all this information.
- And that's just one kind of sensor. Add in other sensors like light, air quality, and temperature, and these algorithms can learn much much more.
- With thousands of farms all collecting this information, these algorithms can create incredible insights into how to make crops grow the best.



INTERNET OF THINGS (IOT)

DEFINITION OF IoT

- 1 The internet of Things is about extending the power of the internet beyond computers and smartphones to a whole range of other things, processes and environments.
- 2 The internet of things (IoT) is a computing concept that describes the idea of everyday physical objects being connected to the internet and being able to identify themselves to other devices.

Most people think about being connected in terms of computers, tablets and smartphones. IoT describes a world where just about anything can be connected and communicate in an intelligent fashion.



WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

0866 244 44 72

98499 42299

94415 27588